```
automatic_flush

//automatic flush using Ultrasonic Sensor HC-SR04 by S.C.DWIVEDI
// defines pins numbers
const int trigPin1 = 2;      //defining echo sensor pins  1
const int echoPin1 = 3;
const int relayPin =  13;      // the number of the buzzer pin
// defines variables
long duration1;          //defining echo sensor varables 1
int distance1;

void setup() {
   pinMode(relayPin, OUTPUT);
pinMode(trigPin1, OUTPUT); // Sets the trigPin as an Output 1
pinMode(echoPin1, INPUT); // Sets the echoPin as an Input   1
Serial.begin(9600); // Starts the serial communication
}

void loop() {
// Clears the trigPin
digitalWrite(trigPin1, LOW);
delayMicroseconds(2);
digitalWrite(trigPin1, HIGH);// Sets the trigPin on HIGH state for 10 micro seconds
delayMicroseconds(10);
digitalWrite(trigPin1, LOW);
duration1 = pulseIn(echoPin1, HIGH);// Reads the echoPin, returns the sound wave travel time in microseconds
distance1= duration1*0.034/2;       // Calculating the distance

if ( distance1 <= 30) {            //logical OR , compare with safe distance
    // turn relay on:
```

Fig. 4: A snapshot of the source code

TABLE 1
CONNECTIONS BETWEEN HC-SR04 AND ARDUINO PINS

Ultrasonic Sensor HC-SR04	Arduino Board Pins
Vcc	Vcc (+5V)
Trig	pin16(D1/TX)
Echo	pin17(D2)
GND	FND

from the sensor, beyond 5cm, it reads the value and toggles the relay back to its normally open position and stops the water flow. Table 1 shows the Arduino Uno's pin connections with the sensor.

Working

First install the Arduino IDE and then upload the source code (see Fig. 4) to the Arduino by selecting the right board and port number in Arduino IDE. The Arduino Uno's code checks the input of ultrasonic sensor reading and based on that reading it gets to know whether someone is within 5cm in front of it or not. It gives the output to the trigger pin and drives the solenoid valve when it finds someone within 5cm.

Normally, the sensor output is low and the valve is not energised. When a person comes in front of the ultrasonic sensor, the sensor's output becomes high and remains high as long as the person is within the range. This keeps the solenoid coil energised so that the valve remains open and the water keeps flowing from the tap.

When the sensor detects that there is no one within its range, its output becomes low and the solenoid coil gets de-energised. Thus, water flows from the tap only when someone is within its range and stops automatically when the person moves away.

Being quite simple, the circuit can be assembled on a general-purpose PCB for testing. After assembling the circuit, enclose it in a suitable box. Install the ultrasonic sensor in front of the handwash sink in such a way that when someone comes in front of it, the tap water starts flowing.

Fix the solenoid valve in proper water flow direction. The direction is marked in the valve housing. There is a minimum pressure required for proper functioning of the solenoid valve, for which the water storage tank should be at a height of at least three metres above the solenoid valve. Otherwise, use of a water pressure pump would be required.

Do not forget to upload the source code (automatic_flush.ino) before assembling the circuit. For uploading the source code use a USB cable and remove it from laptop/computer after uploading the code. Now connect 12V DC from an adaptor or from a 12V battery across CON1 to operate the circuit. EFY

Bonus. You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/live-diy-arduino-based-flash-light*

PARTS LIST

Semiconductors:
- Board1 - Arduino board
- D1 - 1N4007 rectifier diode
- T1 - BD139 npn transistor

Resistor (1/4-watt, ±5% carbon):
- R1 - 220-ohm

Miscellaneous:
- CON1 - 2-pin connector
- CON2 - 4-pin connector
- 12V, 2A power adaptor/12V battery
- 12V normally closed solenoid valve
- Breadboard or PCB
- Jumper wire
- HC-SR04 ultrasonic sensor

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

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